

Supporting Information for:

Lagrangian evolution of phytoplankton patchiness and community structure in a cyclonic Gulf Stream eddy

1 Text S1: Slope Sea and Sargasso Sea mask

Monthly absolute dynamic topography (ADT; i.e. sea surface height with the geoid removed) was retrieved from the Copernicus Marine Service. Monthly chl-a was retrieved from the Ocean Colour Climate Change Initiative (OC-CCI) merged product (Sathyendranath et al., 2019) and monthly Moderate Resolution Imaging Spectroradiometer (MODIS) sea surface temperature at 4 km was retrieved from NASA EarthData between 2019 - 2024. For publication, this will be updated to 2000-2020 to match the temporal scope of chl-a and ADT data, but cluster results were similar with the exclusion of SST. Monthly climatologies were created from the data by averaging by month and regridding onto a common 0.25°by 0.25°grid. Monthly climatological values of ADT, chl-a, and SST were normalized to their z-scores and used as inputs of a K-Means cluster algorithm. Three clusters sufficed for a coarse identification of the Slope Sea and Sargasso Sea. The Slope Sea waters were characterized by low ADT, high chl-a, and cold SST, and Sargasso Sea waters were characterized by high ADT, low chl-a, and warm SST.

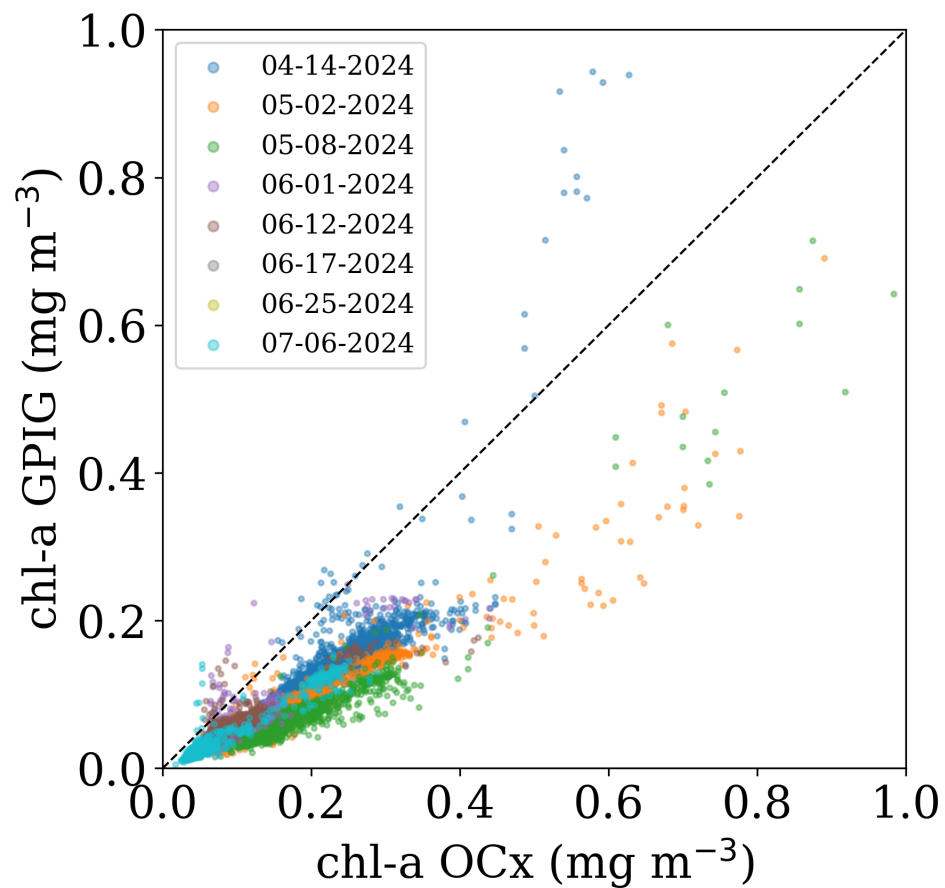


Figure 1: Chlorophyll-a from the GPig algorithm plotted against chlorophyll-a from NASA's OCx algorithm. Different colors represent different dates. 1:1 line in black.

3 Figure S2

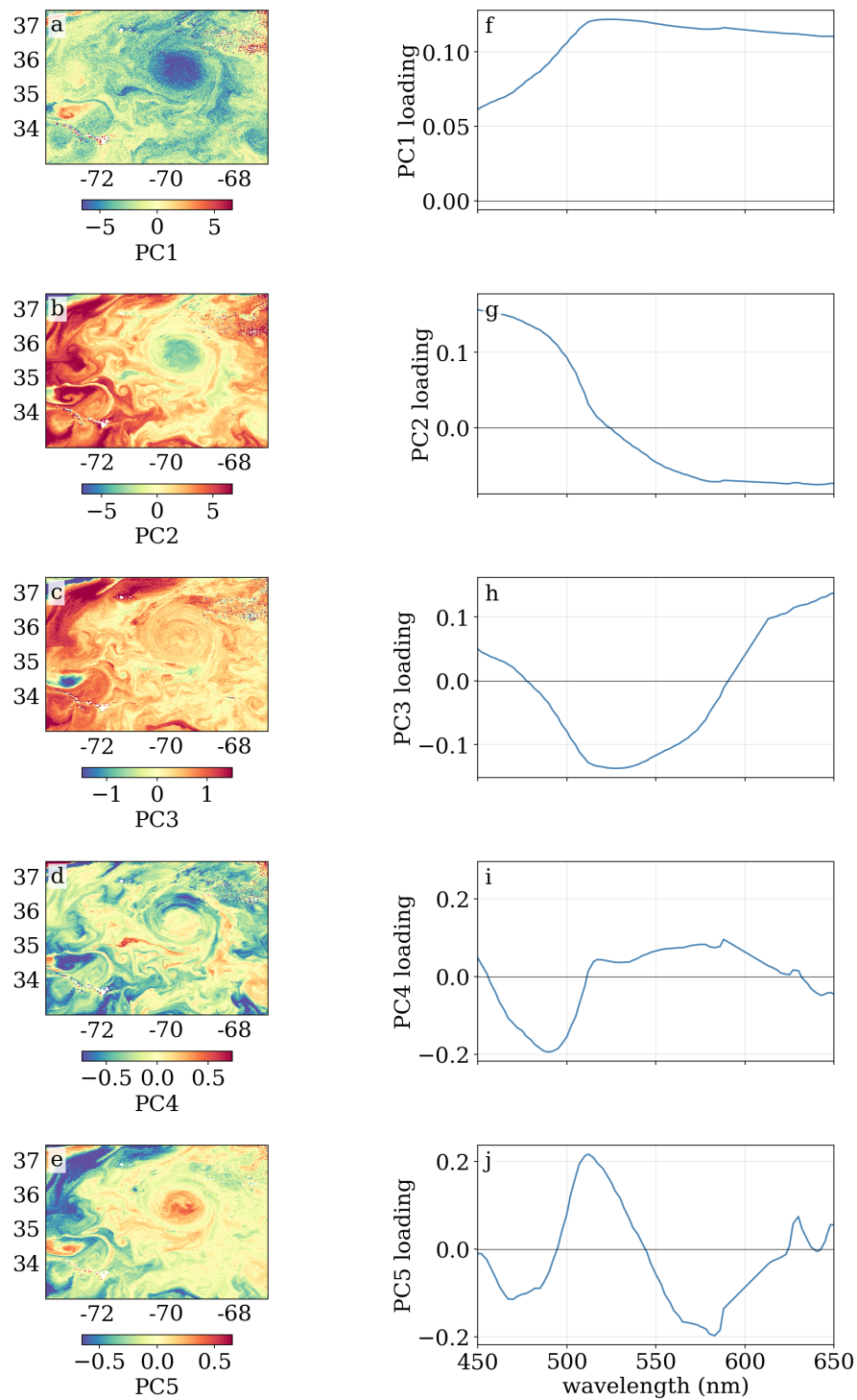


Figure 2: Principal components 1-5 cropped to eddy of interest (a-e). Principal component loadings for PC1-5 plotted against wavelength (f-j).

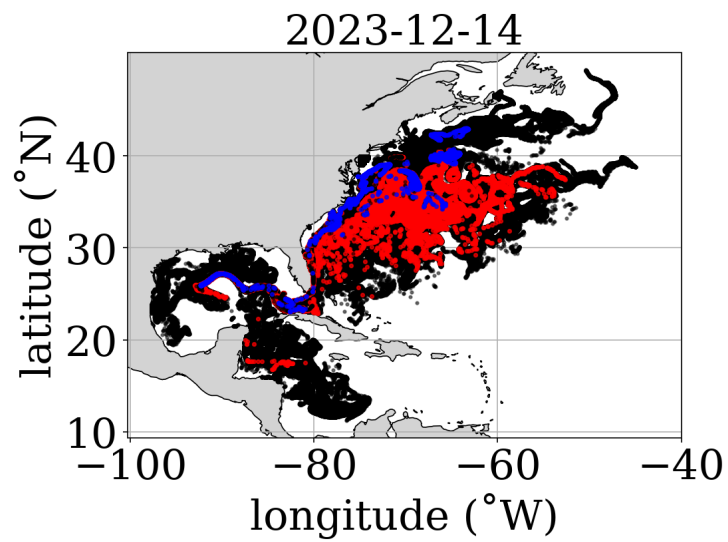


Figure 3: Particle locations from backward particle simulations on December 14, 2023. Particles were seeded on April 14, 2024 and traced backwards in time. Geographic extent shows the U.S. Eastern Seaboard.

5 Figure S4

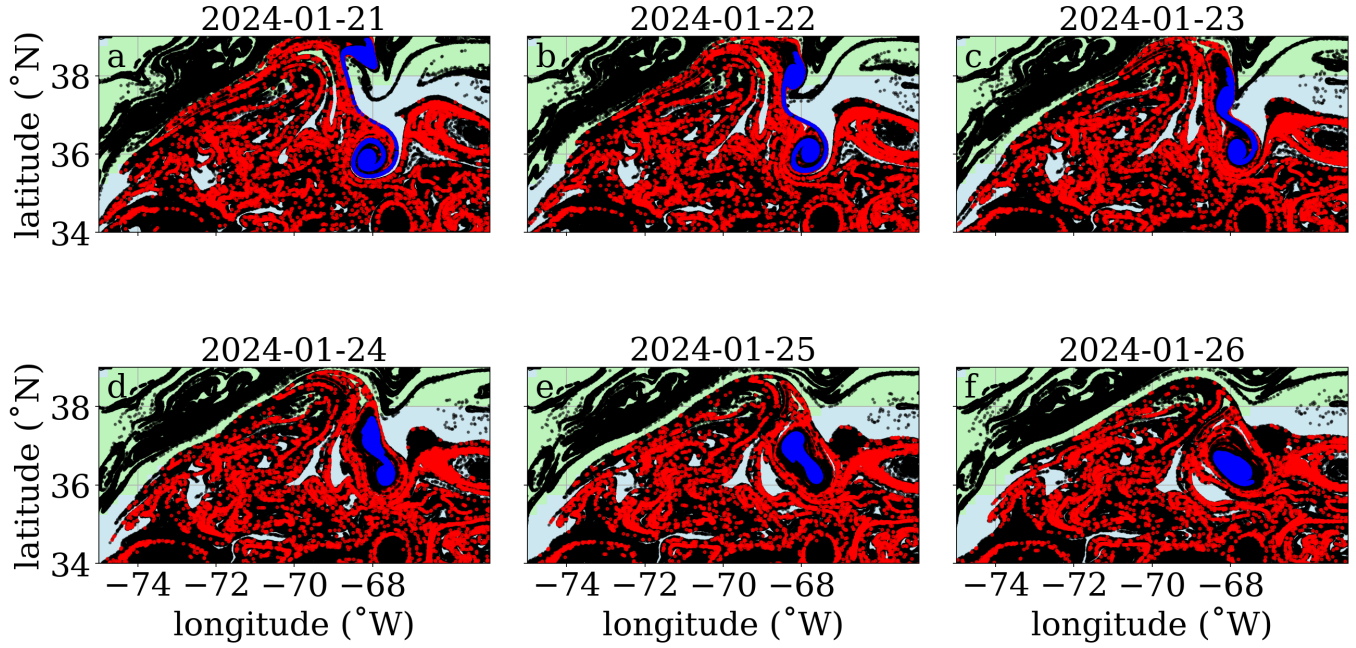


Figure 4: Coalescence of eddy core occurring at the end of January, 2024 (blue). Particle trajectories colored as in Fig. S3. Green denotes Slope Sea Waters and light blue denoted Sargasso Sea Waters. Blue particles denote those sourced from the core and red from the periphery.

6 Figure S5

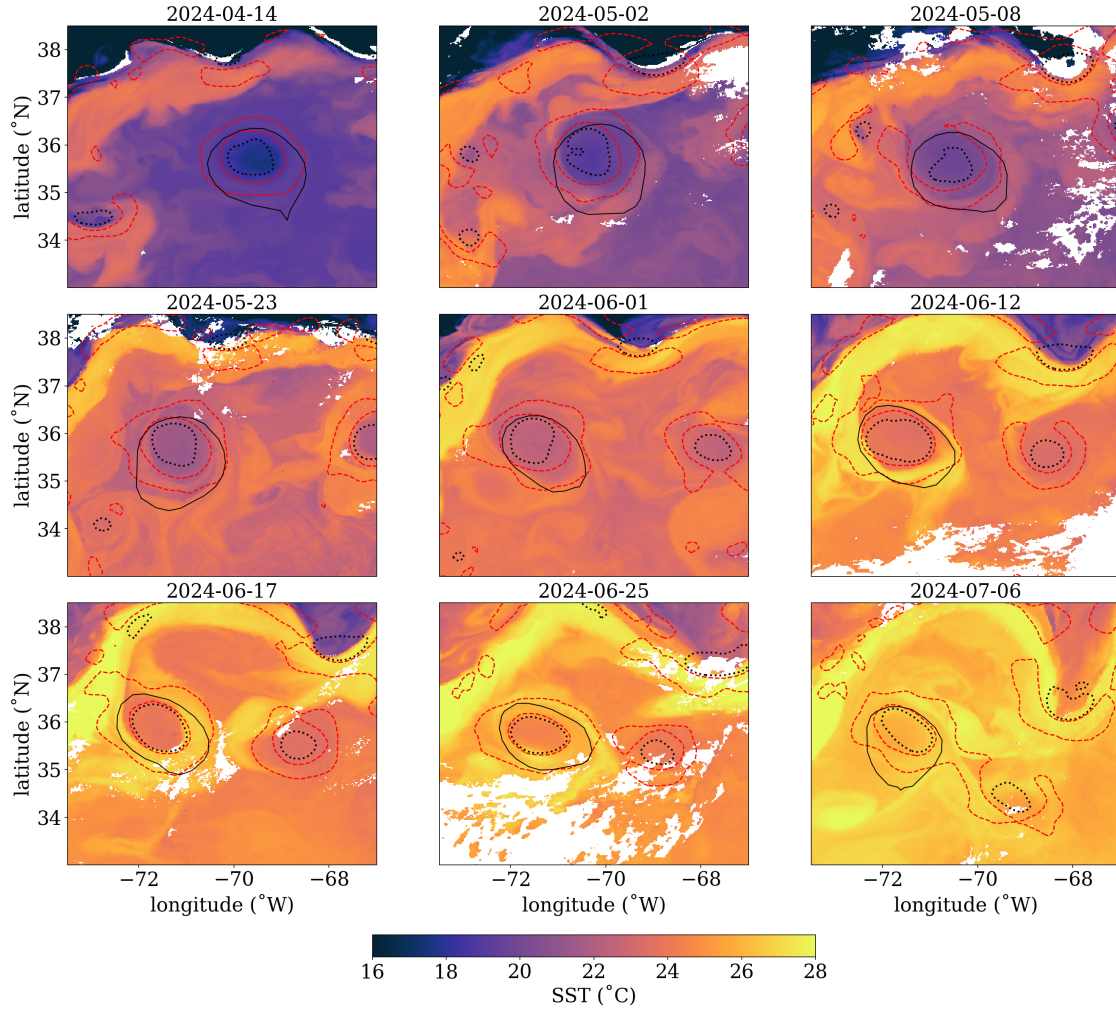


Figure 5: Select snapshots of sea surface temperature from L3S-LEO-DY dataset (0.02° spatial resolution) surrounding the Gulf Stream cold core ring of interest. Red contours denote $0.5 \times 10^{-9} s^{-2}$ and white dotted contours denote $-2 \times 10^{-9} s^{-2}$ Okubo-Weiss thresholds.

7 Figure S6

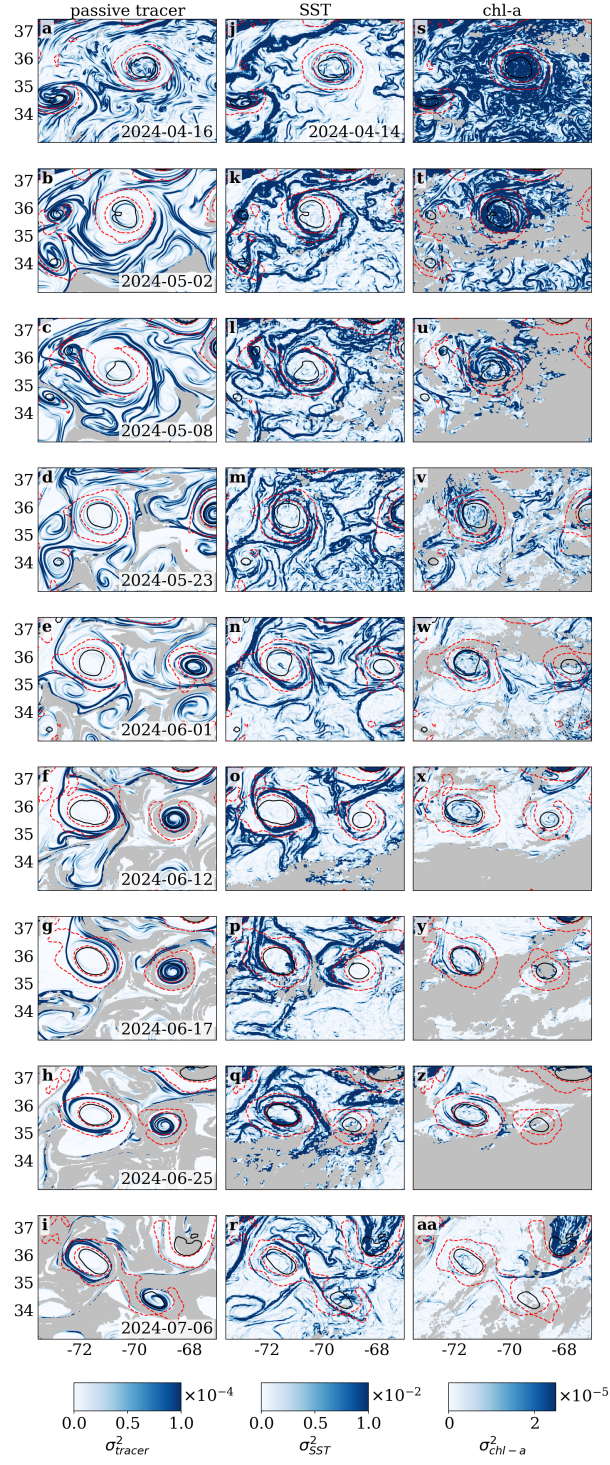


Figure 6: Variance for passive tracer (a-i), SST (j-r), and chl-a (s-aa) for select clear day satellite snapshots. FTLE ridges, detected by contours of $2 \times 10^{-6} s^{-1}$, in red. Gray regions denote no data due to particles leaving the domain (a-i) or cloud cover (j-aa).

8 Figure S7

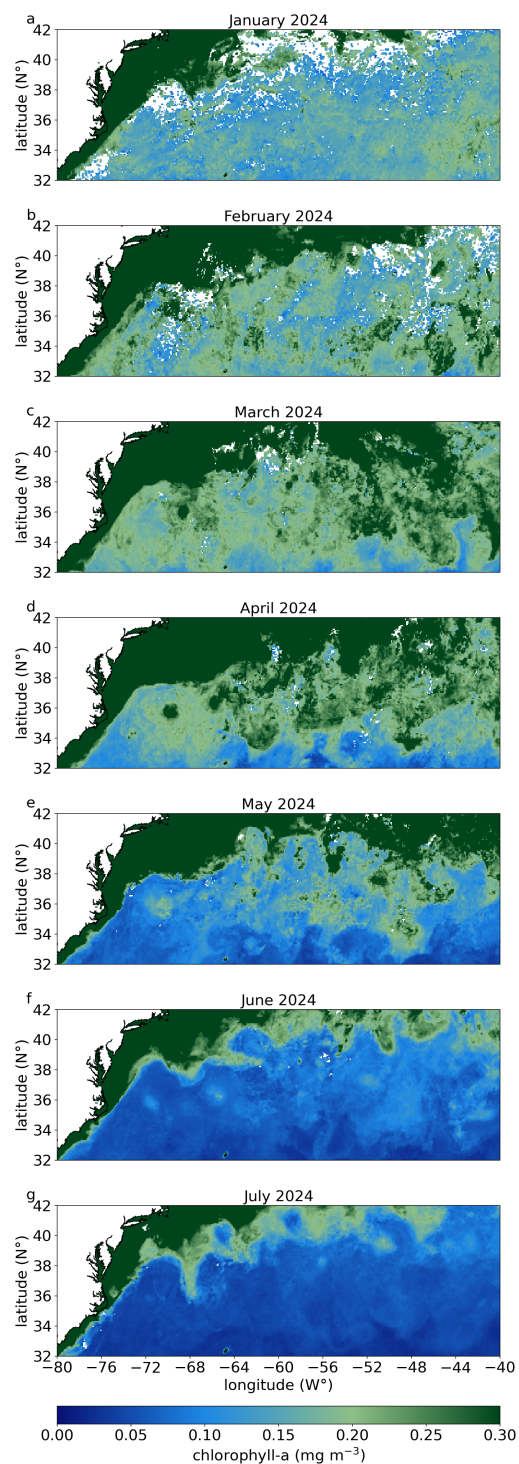


Figure 7: Monthly chlorophyll-a from Sentinel-3A OLCI at a 9km resolution for January - July 2024.

9 **Figure S8**

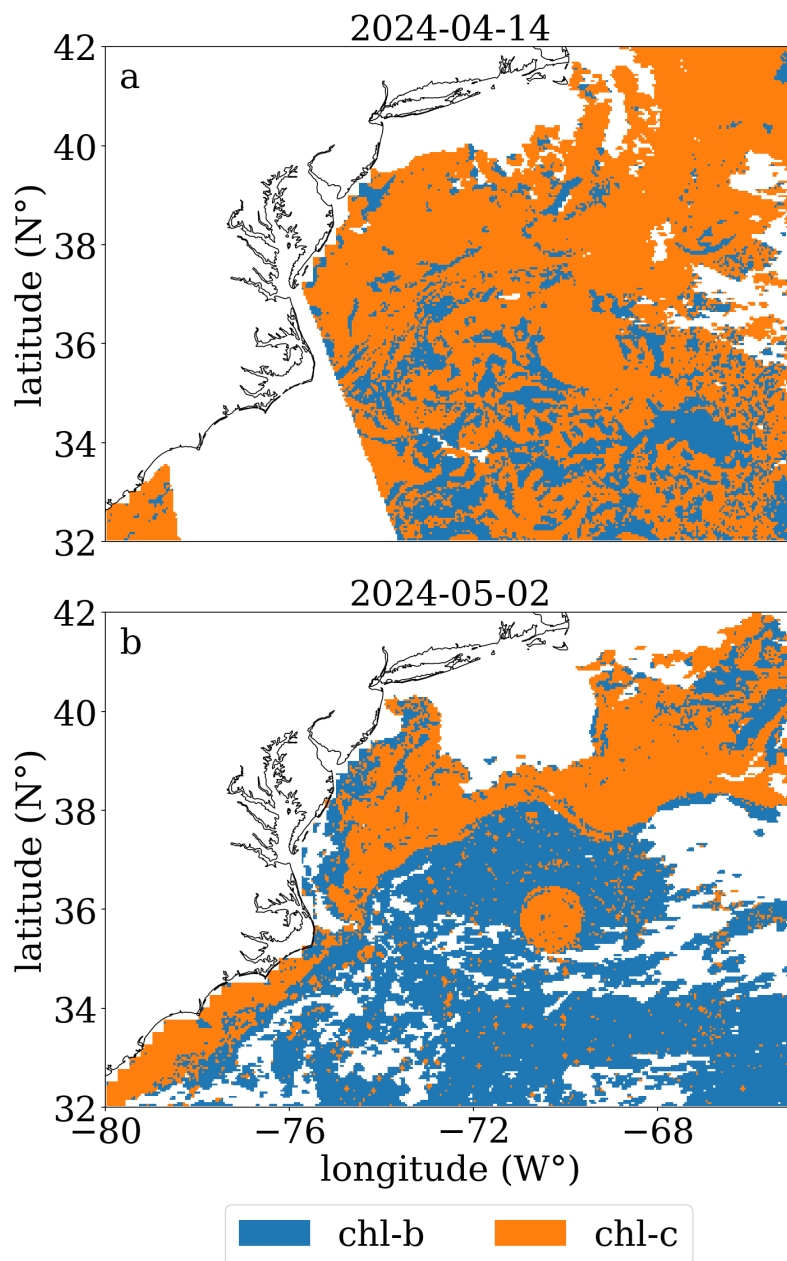


Figure 8: Dominant accessory pigment from chl-b and chl-c yielded from GPig run on Level-3 PACE data on April 14, 2024 and May 2, 2024 (4 km spatial resolution).

10 **Figure S9**

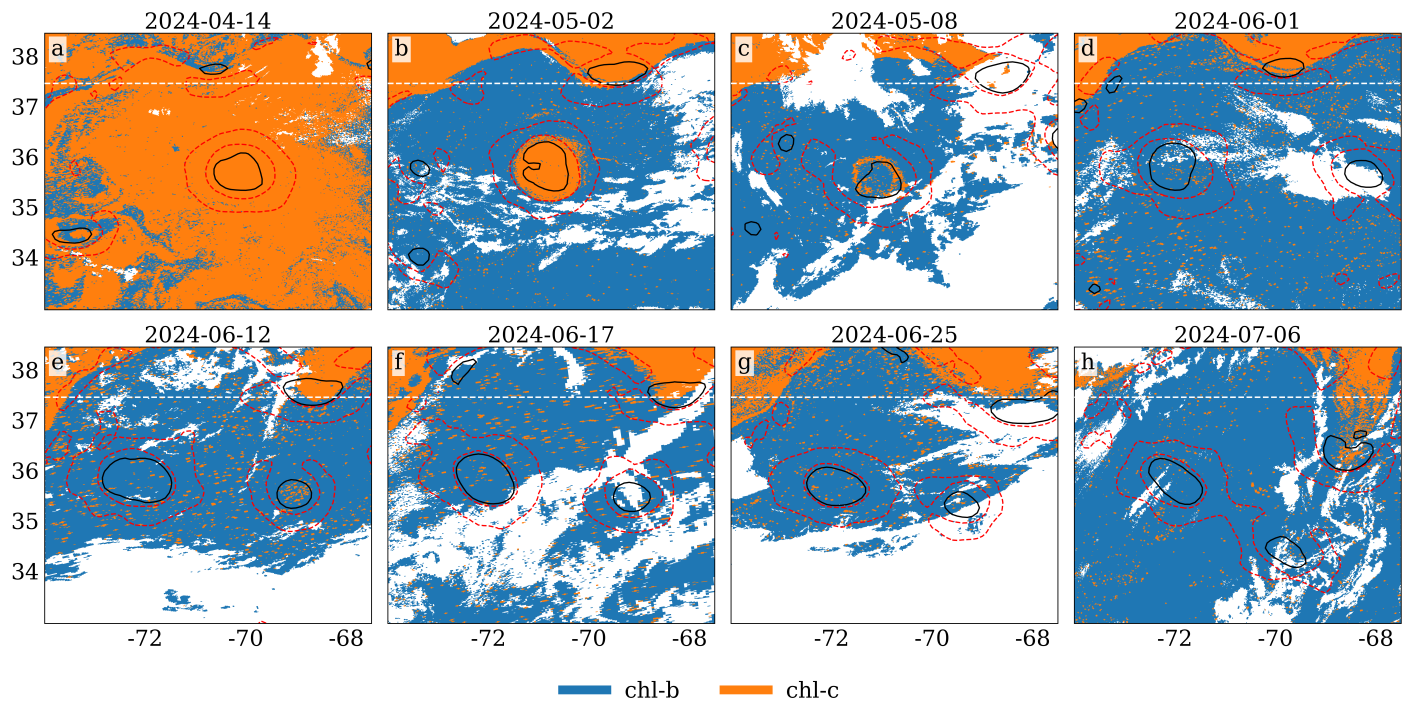


Figure 9: Dominant accessory pigment. White horizontal dashed line denotes spatial extent plotted in Fig. S6. Red contours denote $0.5 \times 10^{-9} s^{-2}$ and white dotted contours denote $-2 \times 10^{-9} s^{-2}$ Okubo-Weiss thresholds.

11 Figure S10

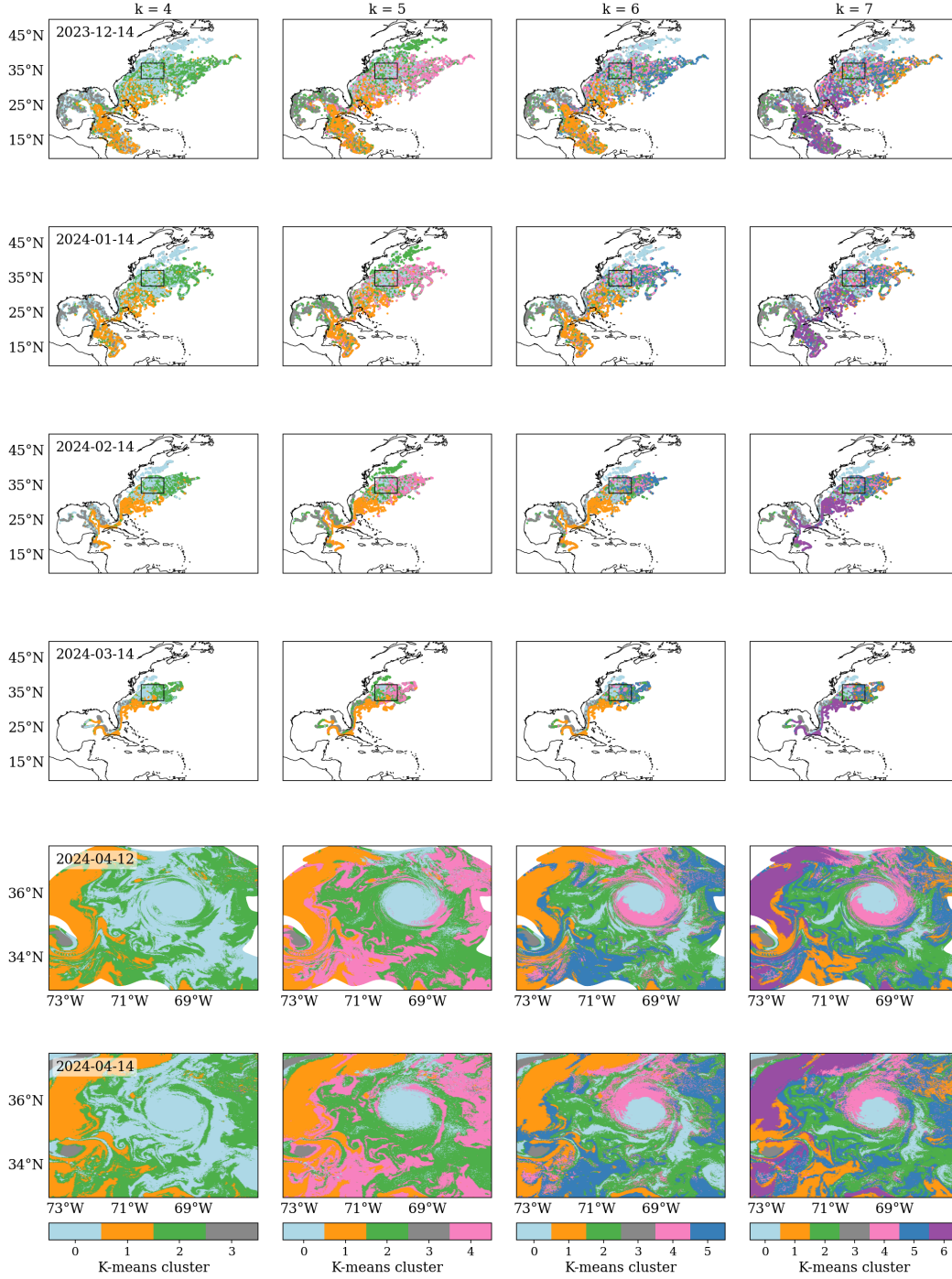


Figure 10: K-Means clustering results run with 4, 5, 6, and 7 clusters (last row). Particles were tracked backward in time to yield their positions on December 14, 2023 (first row), January 14, 2024 (second row), February 14, 2024 (third row), March 14, 2024 (fourth row), and April 12, 2024 (fifth row).

12 Figure S11

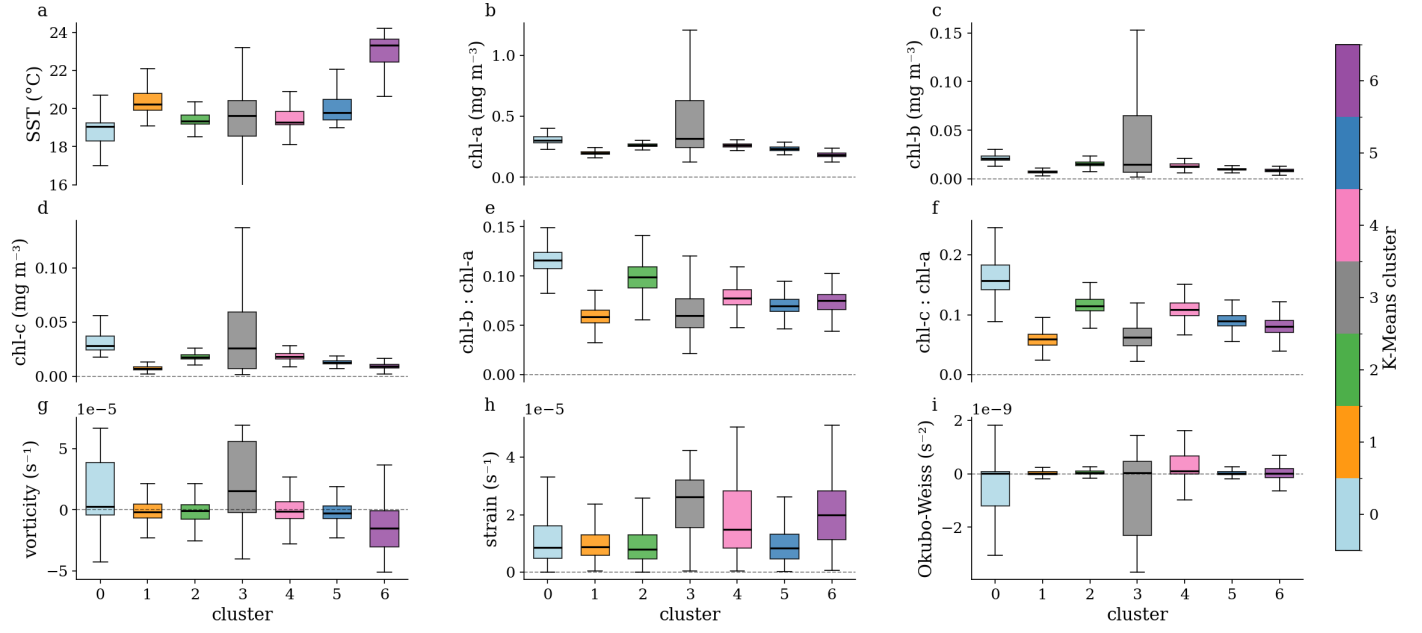


Figure 11: Boxplots for select variables grouped by cluster. Results shown for K-Means run with 7 clusters.

13 References

E.U. Copernicus Marine Service Information (CMEMS): Global Ocean Gridded L4 Sea Surface Heights and Derived Variables Reprocessed 1993 Ongoing, Marine Data Store [Dataset], <https://doi.org/10.48670/moi-00148>, 2024.

Sathyendranath, S., Brewin, R. J. W., Brockmann, C., Brotas, V., Calton, B., Chuprin, A., Cipollini, P., Couto, A. B., Dingle, J., Doerffer, R., Donlon, C., Dowell, M., Farman, A., Grant, M., Groom, S., Horseman, A., Jackson, T., Krasemann, H., Lavender, S., . . . Platt, T. (2019). An Ocean-Colour Time Series for Use in Climate Studies: The Experience of the Ocean-Colour Climate Change Initiative (OC-CCI). *Sensors*, 19(19), 4285. <https://doi.org/10.3390/s19194285>